

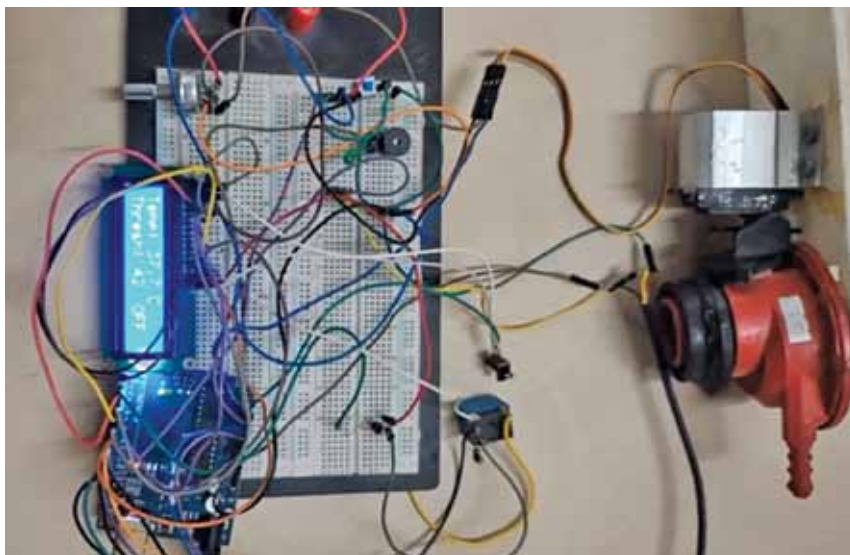


Fig. 3: The author's prototype

Within the `loop()` function, logic is implemented to select the operating mode, read temperature data, and control the servo motor or relay based on measured values. Functionality to accept temperature input through the keypad is also included.

After completion, the correct board and port are selected in the Arduino IDE, followed by compilation and uploading of the program. Fig. 2 shows a snippet of the source code.

**EFY note.** The code may fail to upload if pins D0 (RX) and D1 (TX) are used for the relay, as these pins are shared with USB communication. Uploading should be completed before connecting the circuit, or D1 (TX) should be disconnected during uploading and reconnected afterwards. Adding a delay of about

40 seconds in the `setup()` function helps avoid conflicts.

### Construction and testing

The circuit (see Fig. 1) can be assembled on a breadboard as shown in Fig. 3. The DS18B20 sensor is connected with a 4.7k $\Omega$  pull-up resistor between its data and VCC pins. The 4  $\times$  4 keypad is interfaced with the assigned digital pins.

The 16  $\times$  2 LCD is connected in parallel mode along with a 10k $\Omega$  potentiometer for contrast adjustment. The servo motor is connected to pin A4. The relay is driven using a BC547 or 2N2222 transistor with a 1k $\Omega$  base resistor and a flyback diode across the relay coil.

The buzzer, LED, and slide switch are connected as per the circuit diagram. The AC load (heater

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or immersion rod) is connected through the relay contacts with proper insulation.

Before wiring, the code should be uploaded to the Arduino. The system is powered on, and the desired temperature is set using the keypad. The operating mode is selected using the slide switch: with servo mode used for gas control and relay mode for heater control.

The system monitors temperature continuously and automatically switches off the gas valve or heater once the set value is reached.

Operation is verified by checking the display output, keypad response, and switching action. Fig. 3 shows the working prototype. **EFY**

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